NMC 2015 O-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. Evaluate $\left(\frac{1}{16}\right)a^0 + \left(\frac{1}{16}\right)^0 - 64^{-\frac{1}{2}} - (32)^{-\frac{4}{5}}$.

Qn2. Simplify the following expression $\frac{200012}{200013^2 - 200012 \times 200014}$.

Qn3. After finding the average of 125 scores, a student carelessly included the average with the 125 scores and found the average of these 126 numbers. What is the relationship between the true average and the second average obtained?

Qn4. Let $f(2x-1) = 4x^2 - 10x + 16$. Find $f(x)$.

Qn5. A number M can be written as 47 in base x and as 74 in base y. Assuming that both x and y are positive integers, find the least possible value of $x + y$.

Qn6. Evaluate x^4 where $x = \sqrt{1 + \sqrt{1 + \sqrt{1}}}$.

Qn7. Some chocolates are missing and one of three children ate them. At least one child told the truth and at least one lied. The children made the following statements.
(i) Jonh: *Mary did not eat them.* (ii) Mary: *Either John or Tom ate them.*
(iii) Tom: *John ate them.*
Who actually ate the chocolates?

Qn8. Without using a calculator, evaluate $G(x) = \frac{1}{\sqrt{\frac{3}{2} + \frac{2}{\sqrt{2}}}} + \frac{1}{\sqrt{\frac{3}{2} - \frac{2}{\sqrt{2}}}}$.

Qn9. The 3−digit number $2a3$ is added to the number 326 to give the 3−digit number $5b9$. Given that $5b9$ is divisible by 9, what is the value of $a + b$.

Qn10. The number of boys that wrote a mathematics contest in 2014 was 20% higher than in 2013. The number of girls that wrote the contest in 2014 had increased by 50% over the same period, while the total number of contestants had increased by only 30%. What fraction of the students that wrote the contest in 2014 were girls?

SECTION B

Qn11. The lengths of the sides of an acute angled triangle are successive integers. Show that the altitude to the second longest side divides it into two segments whose difference in length equals 4.

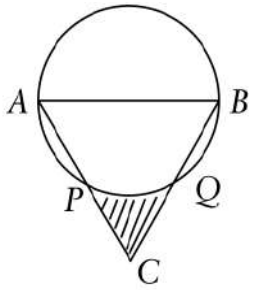
Qn12. A list of five positive integers has mean 12 and range 18. The mode and the median are both 8. Find the largest possible integer in the list?

Qn13. (a) Let $\log_a b = \log_b a$ where $a \neq b$, $ab > 0, a \neq 1$ and $b \neq 1$. Find the value of ab.

(b) John walked a certain distance at a constant rate. If he had gone $\frac{1}{2}$ km/hr faster, he would have walked the distance in four-fifths of the time. If he had gone $\frac{1}{2}$ km/hr slower, he would have been $2\frac{1}{2}$ hours longer on the road. Find the distance travelled?

Qn14. Student lockers at a university are numbered consecutively beginning with locker number 1. The plastic digits used to number the lockers cost shs.200 each. If it costed shs.$1,390,600$ to label all the lockers, how many lockers are there at the university?

Qn15. In the diagram below, ABC is an equilateral triangle whose base AB is a diameter of the circle with $AB = 12\ cm$ and AC, BC intersect the circle at points P and Q respectively as shown. Find the area of the shaded region.



NMC 2015 O-LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1. $\left(\frac{1}{16}\right) a^0 + \left(\frac{1}{16}\right)^0 - 64^{-\frac{1}{2}} - (32)^{-\frac{4}{5}} = \frac{1}{16} + 1 - \frac{1}{\sqrt{64}} - \frac{1}{(\sqrt[5]{32})^4} = \frac{1}{16} + 1 - \frac{1}{8} - \frac{1}{16} = 1 - \frac{1}{8} = \frac{7}{8}$.

Qn2. Let $200012 = n$. Then $\frac{200012}{200013^2 - 200012 \times 200014} = \frac{n}{(n+1)^2 - n(n+2)} = \frac{n}{n^2+2n+1-n^2-2n} = n = 200012$.

Qn3. True average, $x = \frac{x_1 + x_x + x_3 + \cdots + x_{125}}{125}$.

Fake average $y = \frac{x_1 + x_x + x_3 + \cdots + x_{125} + x}{126} = \frac{125x + x}{126} = \frac{126x}{126} = x$.

Therefore the true average and the fake average are the same.

Qn4. Let $g(x) = f(2x-1) = 4x^2 - 10x + 16$. Then $g(\frac{x+1}{2}) = f(\frac{2(x+1)}{2} - 1) = f(x)$. Hence

$$\begin{aligned} f(x) &= 4\left(\frac{x+1}{2}\right)^2 - 10\left(\frac{x+1}{2}\right) + 16 \\ &= (x+1)^2 - 5(x+1) + 16 \\ &= x^2 + 2x + 1 - 5x - 5 + 16 \\ &= x^2 - 3x + 12. \end{aligned}$$

Therefore the function $f(x) = x^2 - 3x + 12$.

Qn5. $(47)_x = (74)_y$ implies that $4x + 7 = 7y + 4$. So, $7y - 4x = 3 \Longrightarrow x = \frac{7y-3}{4}$.
We note that both x and y are integers each greater than 7. Suppose $y = 8$, then $x = \frac{56-3}{4} = \frac{53}{4}$ which is not an integer. Suppose $y = 9$, then $x = \frac{63-3}{4} = \frac{60}{4} = 15$.

The least value of $x + y = 15 + 9 = 24$.

Qn6. Evaluate x^4 where $x = \sqrt{1 + \sqrt{1 + \sqrt{1}}}$. Since $\sqrt{1} = 1$, then $x = \sqrt{1 + \sqrt{2}}$.

$x^2 = \left(\sqrt{1 + \sqrt{2}}\right)^2 = 1 + \sqrt{2}$. Hence $x^4 = (x^2)^2 = (1 + \sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$.

Qn7. Suppose John ate the chocolates, then all the three told the truth, a contradiction. If Mary ate the chocolates, then all the three told lies which is also a contradiction. If Tom ate the chocolates, then John and Mary told the truth and Tom lied. Hence Tom ate the chocolates.

Qn8.

$$G(x) = \frac{1}{\sqrt{\frac{3}{2} + \frac{2}{\sqrt{2}}}} + \frac{1}{\sqrt{\frac{3}{2} - \frac{2}{\sqrt{2}}}} = \frac{\sqrt{\frac{3}{2} - \frac{2}{\sqrt{2}}} + \sqrt{\frac{3}{2} + \frac{2}{\sqrt{2}}}}{\sqrt{\left(\frac{3}{2} + \frac{2}{\sqrt{2}}\right)\left(\frac{3}{2} - \frac{2}{\sqrt{2}}\right)}}$$

Let $a = \sqrt{\frac{3}{2} + \frac{2}{\sqrt{2}}}$ and $b = \sqrt{\frac{3}{2} - \frac{2}{\sqrt{2}}}$ Then $(a + b)^2 = a^2 + b^2 + 2ab = \frac{3}{2} - \frac{2}{\sqrt{2}} + \frac{3}{2} + \frac{2}{\sqrt{2}} +$

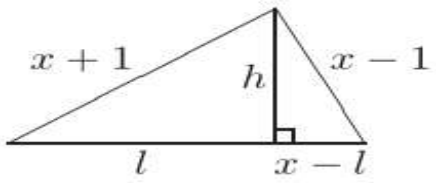
$2\sqrt{\left(\frac{9}{4} - \frac{4}{2}\right)} = 3 + 2 \times \frac{1}{2} = 4$. Therefore $\frac{\sqrt{\frac{3}{2} - \frac{2}{\sqrt{2}}} + \sqrt{\frac{3}{2} + \frac{2}{\sqrt{2}}}}{\sqrt{\left(\frac{3}{2} + \frac{2}{\sqrt{2}}\right)\left(\frac{3}{2} - \frac{2}{\sqrt{2}}\right)}} = \frac{\sqrt{4}}{0.5} = \frac{2}{0.5} = 4$.

Qn9. Since $5b9$ is divisible by 9, then the sum $5 + b + 9 = 14 + b$ is also a multiple of 9. Thus the digit $b = 4$. Now, $2a3 = 5b9 - 326 = 549 - 326 = 223$. This implies that $a = 2$. Hence $a + b = 4 + 2 = 6$.

Qn10. Let x and y be the numbers of boys and girls, respectively, that wrote the contest in 2013. Then the corresponding numbers for the year 2014 are $1.2x$ and $1.5y$. The total number of contestants in 2014 are $1.3(x + y)$. We solve $1.2x + 1.5y = 1.3(x + y)$ to get $0.2y = 0.1x$ or $x = 2y$. The required fraction is $\frac{1.5y}{1.3(x+y)} = \frac{1.5y}{1.3(3y)} = \frac{1.5}{3.9} = \frac{5}{13}$.

SECTION B

Qn11. Let the lengths of the triangle be $x - 1, x$ and $x + 1$ and the length of the perpendicular projection be h and the length of the longer section of the side of length x be l as shown.



Therefore

$$\begin{aligned} h^2 = (x+1)^2 - l^2 &= (x-1)^2 - (x-l)^2 \\ x^2 + 2x + 1 - l^2 &= x^2 - 2x + 1 - x^2 + 2lx - l^2 \\ 4x + x^2 &= 2lx \\ 4 + x &= 2l. \\ l - (x-l) &= 2l - x = 4. \end{aligned}$$

Therefore the altitude to the second longest side divides it into two segments whose difference in length equals 4.

Qn12. Let the numbers be x_1, x_2, x_3, x_4, x_5 in ascending order. Clearly $x_3 = 8$ and $x_5 = 18 + x_1$. Therefore the numbers are $x_1, x_2, 8, x_4, x_1+18$. The mean is 12 which implies that $x_1+x_2+8+x_4+x_1+18 = 60$ or $2x_1 + x_2 + x_4 = 34$. Since the mode is 8, it is possible that $x_2 = 8, x_4 = 8$ or both. Suppose $x_2 = x_4 = 8$, then $2x_1 = 18 \Longrightarrow x_1 = 9$. This is impossible given that x_1 is the smallest so either is 8 but not both.

Suppose that $x_4 = 8$, then $2x_1 + x_2 = 26$ and $x_1 \leq x_2 < 8$. There are no possible pairs (x_1, x_2) satisfying the two so $x_2 = 8$.

In this case $2x_1 + x_4 = 26$ and $x_1 \leq 8 < x_4 < x_1 + 18$. The largest possible value of x_1 is 8 in which the largest possible integer is $8 + 18 = 26$.

We only check that $x_4 = 26 - 16$ does not violate $8 < x_4 < x_1 = 18$ which is the case.

Qn13. (a) Let $\log_a b = \log_b a$. This implies that $b = a^x$ and $a = b^x$. Multiplying the two equations, we get $ab = (ab)^x$. There are three possibilities.

(i) $ab = 0$ (ii) $x = 1$ making $a = b$ (iii) $ab = 1$.

The first two are excluded by the conditions $a \neq b$ and $ab > 0$. We remain with the third possibility that $ab = 1$.

(b) All the three walking rates and the corresponding times yield the same distance. Let R, T denote the first rate and the corresponding time, then we have the following distance expressions.

$$RT = (R + \frac{1}{2})(\frac{4}{5})T) \tag{1}$$

$$RT = (R - \frac{1}{2})(T + \frac{5}{2}) \tag{2}$$

From (1), $RT - \frac{4}{5}RT = \frac{4}{10}T$. Hence $\frac{1}{5}RT = \frac{2}{5}T \Longrightarrow RT = 2T$.

From Equation (2), $RT = (2 - \frac{1}{2})(\frac{RT}{2} - \frac{5}{2}) = \frac{3}{2}(\frac{RT+5}{2}) = \frac{3}{4}(RT + 5)$.
Thus $RT = 15$. The distance travelled is 15km.

Qn14. The locker labelling requires $\frac{1390600}{200} = 6953$. Lockers 1 through 9 require 9 digits. Lockers 10 through 99 require $2 \times 90 = 180$ digits. Lockers 100 through 999 require $3 \times 900 = 2700$ digits. Hence the remaining lockers require $6953 - 2700 - 180 - 9 = 4064$ digits, and so there must be $\frac{4064}{4} = 1016$ more lockers each using 4 digits. In all, there are $1016 + 999 = 2015$ student lockers.

Qn15. Let N be the centre of the circle and then draw the two radii NP and NQ so that $AN = NB = NP = NQ = 6cm$. $\triangle ABC$ is equilateral so $< PAN = 60^\circ$ and hence $< APN = 60^\circ$ since $AN = NP$. It follows that $< ANP = 60^\circ$ so, $\triangle ANP$ is equilateral.

Similarly, $< QNB = 60^\circ$ and therefore $< PNQ = 60^\circ$ as well. Let $\bigotimes$ be the area of the figure bounded by the points. Then

$$\begin{aligned} \otimes ABC &= \otimes ANP + \otimes BNQ + \otimes NPQ + \otimes PQC \\ \otimes PQC &= \otimes ABC - (\otimes ANP + \otimes BNQ + \otimes NPQ) \end{aligned}$$

$$\text{But} \otimes ANP = \frac{1}{2} \times a \times b \times \sin\theta = \frac{1}{2} \times 6 \times 6 \times \sin 60^\circ = 9\sqrt{3}.$$

$$\text{Similary} \otimes BNQ = \frac{1}{2} \times a \times b \times \sin\theta = \frac{1}{2} \times 6 \times 6 \times \sin 60^\circ = 9\sqrt{3}.$$

$$\text{Also} \otimes NPQ = \frac{\theta}{360} \times \pi r^2 = \frac{60}{360} \times \pi \times 6^2 = 6\pi.$$

$$\otimes ABC = \frac{1}{2} \times 12 \times 12 \times \sin 60^\circ = \frac{72\sqrt{3}}{2} = 36\sqrt{3}.$$

$$\text{Hence} \otimes PQC = 36\sqrt{3} - 9\sqrt{3} - 9\sqrt{3} - 6\pi = 18\sqrt{3} - 6\pi = 6(3\sqrt{3} - \pi)cm^2.$$

NMC 2015 O-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. For what value of k is the displayed factoring valid?

$$2015^3 + 2015^4 = k(1 + 2015).$$

Qn2. When the decimal point of a certain positive decimal number is moved four places to the right, the new number is nine times the reciprocal of the original number. What is the original number?